

PROJECT CONCEPT NOTE

1. Project Title

VibeChef – AI-Powered Recipe Generator

Application Name - VibeChef

Live Application URL -

[VibeChef | AI Recipe Maker](#)

GitHub Repository –

<https://github.com/kuramvasini8-lab/VibeChef>

2. Problem Statement / Objective

Many people have ingredients available at home but are unsure about what they can prepare with them. Deciding what to cook can become difficult when users have limited ingredients or want to make use of the ingredients they already have.

VibeChef is an AI-powered recipe generation web application designed to help users discover suitable recipes based on the ingredients they have available. The user can enter **all the ingredients they currently have**, and the application uses AI to generate a recipe suggestion based on the provided information.

The main objective of VibeChef is to make recipe discovery easier, faster, and more personalized while helping users make better use of the ingredients available to them.

3. Target Users and Use Case

Target Users

VibeChef is designed for:

- Students and young professionals who want quick recipe ideas.
- Individuals who are unsure about what to cook with available ingredients.
- Home cooks looking for new meal ideas.
- Users who want to make use of ingredients already available in their kitchen.

Use Case

A user opens VibeChef and provides a complete list of the ingredients they have available. The AI processes the provided ingredients and generates a suitable recipe suggestion.

For example, if a user has ingredients such as vegetables, rice, and basic cooking ingredients, they can enter these ingredients into VibeChef to receive a recipe idea based on their available ingredients.

The user should provide **all the ingredients they have available** so that the AI can generate a more relevant recipe suggestion.

4. LLM Model and API Used

VibeChef uses an external Large Language Model (LLM) API to provide AI-powered recipe generation.

AI Technology

- **LLM API:** Groq API
- **Model:** Llama 3.3 70B Versatile
- **Backend Integration:** The LLM API is accessed through the backend server.
- **Security:** The API key is stored as an environment variable and is not exposed directly in the frontend code.

The AI model processes the user's provided ingredient information and generates an appropriate recipe response.

5. Key Features of the Application

5.1 Ingredient-Based Recipe Generation

The core feature of VibeChef is AI-powered recipe generation based on the ingredients provided by the user.

Users can enter **all the ingredients they currently have**, and VibeChef uses the AI model to generate a recipe suggestion based on the provided ingredients.

5.2 AI-Powered Recipe Suggestions

The application uses an LLM to understand the user's ingredient input and generate a relevant recipe response instead of relying only on a fixed list of predefined recipes.

5.3 Simple User Interface

VibeChef provides a simple and easy-to-use interface where users can provide their available ingredients and receive an AI-generated recipe.

5.4 Dynamic AI Response

The recipe content is generated dynamically using the integrated AI model based on the user's input.

5.5 Secure API Integration

The AI API key is kept securely on the server side using environment variables rather than being directly exposed in the frontend application.

5.6 Publicly Accessible Web Application

VibeChef has been deployed online using **Render**, making the application publicly accessible through a live web URL.

6. Expected User Experience and Outcomes

The expected user experience is simple and straightforward:

1. The user opens the VibeChef web application.
2. The user enters **all the ingredients available to them**.
3. The application sends the user's input to the backend.
4. The backend securely communicates with the AI model.
5. The AI processes the provided ingredients.
6. VibeChef presents an AI-generated recipe suggestion to the user.

The expected outcome is to help users quickly decide what they can cook using the ingredients they already have. The application aims to reduce the difficulty of deciding what to prepare and provide users with convenient AI-powered recipe ideas.

7. Technology Stack

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Node.js
Server Framework	Express.js
AI Integration	Groq API
AI Model	Llama 3.3 70B Versatile
Environment Configuration	.env environment variables
Deployment	Render
Containerization	Docker

8. Live Application

The VibeChef application is publicly deployed and accessible through the following link:

[VibeChef | AI Recipe Maker](#)

The live application can be accessed by users to interact with the AI-powered recipe generation functionality.